MASTERVOLT

USERS MANUAL

POWERCHARGER

12/40-3

FULLY AUTOMATIC BATTERY CHARGER

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1 GENERAL INFORMATION

1.1 USE OF THIS MANUAL

This manual contains important safety and operating instructions for the safe and effective operation, maintenance and possible correction of minor malfunctions of the PowerCharger.

It is therefore essential that every person who works on or with the PowerCharger is completely familiar with the contents of this manual, and that he/she carefully follows the instructions and important safety recommendations contained herein. The English version has 20 pages.
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1.2 APPLICABILITY OF THIS MANUAL

All of the specifications, provisions and instructions contained in this manual apply solely to standard versions of the PowerCharger delivered by Mastervolt.
This manual is valid for the following model:

UPC code	Part number	Model
852968002301	44210400	PowerCharger 12/40-3

This model is referred to as the “PowerCharger” throughout this manual.

1.3 USE OF PICTOGRAMS

Safety instructions and warnings are noted in this manual by the following pictograms:

WARNING
A WARNING refers to possible injury to the user or significant material damage to the charger if the user does not (carefully) follow the procedures.

CAUTION!
Special data, restrictions and rules with regard to preventing damage.

A procedure, circumstance, etc which deserves extra attention.

1.4 IDENTIFICATION LABELS

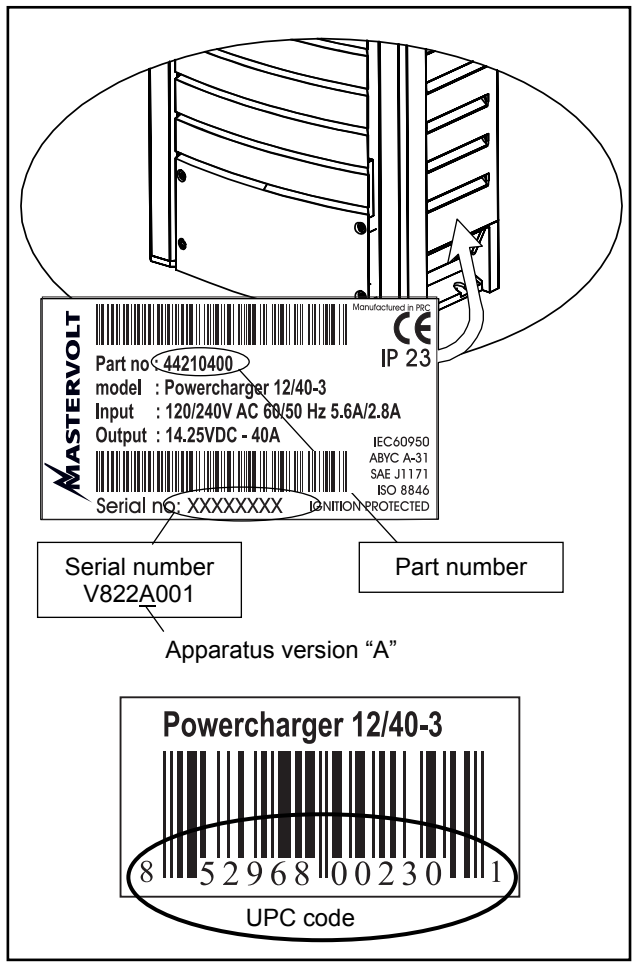


Figure 1: Identification labels

The general identification label is located on the right-hand side of the PowerCharger (see figure 1). Important technical information required for service, maintenance & replacement of parts can be obtained from the identification label.
The UPC code label (figure 1, bottom) is located at the bottom of the PowerCharger.

CAUTION!
Never remove the identification labels.

1.5 LIABILITY

Mastervolt cannot be held liable for:

- consequential damage due to use of the PowerCharger.
- possible errors in the manuals and the results thereof.

2 IMPORTANT SAFETY INSTRUCTIONS

READ AND SAVE THESE INSTRUCTIONS

WARNING
This chapter describes important safety and operating instructions for the use of the PowerCharger in residential, recreational vehicle (RV) and marine applications.

2.1 GENERAL

1 Before using the PowerCharger, read all instructions and cautionary markings on the PowerCharger, the batteries, and all appropriate sections of the manual.
2 To reduce the risk of electric shock – Do not expose PowerCharger to rain, snow, spray, moisture, excessive pollution and extreme temperature swings. To reduce risk of fire hazard, do not cover or obstruct the ventilation openings. Do not install the PowerCharger in a non-ventilated space, overheating may result.
3 Use of an attachment or spare part not recommended or sold by Mastervolt may result in a risk of fire, electric shock, or injury to persons.
4 The PowerCharger is designed to be permanently connected to an AC and DC electrical system. Installation of, and work on the PowerCharger, should be carried out by a qualified, authorized and trained technician or electrician, consistent with the locally applicable standards and regulations.
5 Make sure that all wiring is properly installed and in good electrical condition; and that wire size is large enough for AC ampere rating of the PowerCharger. Check the wiring on a regular basis, at least once a year. Do not use the PowerCharger when the wiring is undersized or damaged.
6 Do not operate PowerCharger if it has received a sharp blow, been dropped, or otherwise damaged in any way; take it to an authorized Mastervolt service center.
7 Except for the connection compartment, see chapter 4, the PowerCharger may not be opened or disassembled. There are no serviceable parts inside the cabinet. Take it to a qualified, authorized and trained technician when service or repair is required. Incorrect reassembly may result in a risk of electric shock or fire. Only qualified, electrician installers are authorized to open the connection compartment.
8 To reduce risk of electric shock, disconnect the PowerCharger from both AC and DC electrical system before attempting any maintenance or cleaning. Turning off controls will not reduce this risk.
9 The PowerCharger must be provided with an equipment-grounding conductor to the AC-input ground terminal. Grounding and all other wiring must comply with local codes and ordinances.
10 Short circuiting or reversing polarity will lead to serious damage to batteries, PowerCharger, wiring as well as accessories. Fuses can not prevent damage caused by reversed polarity and the warranty will be void.
11 In case of fire, you must use the fire extinguisher which is appropriate for electrical equipment.
12 If installed in a marine application external connections to the PowerCharger shall comply with the standards of the American Boat & Yacht Council (ABYC).
13 If installed in a recreational vehicle (RV), external connections to the PowerCharger shall comply with the standards of the Recreational Vehicle Industry Association (RVIA)

2.2 EXPLOSIVE GASES

1 WARNING – RISK OF EXPLOSIVE GASES. WORKING IN VICINITY OF A LEAD-ACID BATTERY IS DANGEROUS. BATTERIES GENERATE EXPLOSIVE GASES DURING NORMAL BATTERY OPERATION. FOR THIS REASON, IT IS OF UTMOST IMPORTANCE THAT EACH TIME BEFORE USING THE POWERCHARGER, YOU READ THIS MANUAL AND FOLLOW THE INSTRUCTIONS EXACTLY.
2 To reduce risk of a battery explosion, follow these instructions and those published by the battery manufacturer and the manufacturer of any equipment you intend to use in vicinity of the battery. Review cautionary markings on these products.
3 The PowerCharger has been rated ignition protected according to SAE J1171 / ISO 8846, only when installed according to this manual and in normal operation! DANGER: To reduce the risk of explosion DURING INSTALLATION OF THE POWERCHARGER - avoid situations where there is danger t explosion

2.3 WARNINGS REGARDING THE USE OF BATTERIES

1 Someone should be within range of your voice or close enough to come to your aid when you are working near a lead-acid battery.
2 Have plenty of fresh water and soap nearby in case battery acid contacts skin, clothing, or eyes.
3 Wear complete eye protection and clothing protection. Avoid touching eyes while working near a battery.
4 If battery acid contacts skin or clothing, wash immediately with soap and water. If acid enters eye, immediately flood eye with cold running water for at least 10 minutes and get medical attention immediately.
5 NEVER smoke or allow a spark or flame in vicinity of a battery or engine.
6 Do not short circuit batteries, as this may result in an explosion and fire hazard! Be extra cautious to reduce the risk of dropping a metal tool onto a battery. It might spark or short-circuit the battery or other electrical part that may cause an explosion.
7 Remove personal metal items such as rings, bracelets, necklaces, and watches when working with a battery. A battery can produce a short-circuit current high enough to weld metallic items like jewellery to metal, causing a severe burn.
8 Only use the PowerCharger for charging the following battery types: Lead Acid/ Wet , AGM, Spiral, Gel or Deep cycle batteries. Do not use PowerCharger for charging dry-cell batteries that are commonly used with home appliances. These batteries may burst and cause injury to persons and damage to property.
9 NEVER charge a frozen battery.
10 Excessive battery discharge and/or high charging voltages can cause serious damage to batteries. Do not exceed the recommended discharge limits of your batteries.
11 If it is necessary to remove a battery, always remove the grounded terminal from battery first. Make sure all accessories are in the OFF position to prevent an arc.
12 Be sure that the area around the battery is well ventilated while the battery is being charged. Refer to the recommendations of the battery manufacturer.
13 Batteries are heavy! It may become a projectile if it is involved in an accident! Ensure adequate and secure mounting and always use suitable handling equipment for transportation.

2.4 WARNING REGARDING LIFE SUPPORT APPLICATIONS

Do not use the PowerCharger for applications in any medical equipment intended for use as a component of any life support system!

2.5 WARRANTY SPECIFICATIONS

Mastervolt guarantees that this unit has been built according to the legally applicable standards and specifications. If modifications take place which are not in accordance with the guidelines, instructions and specifications contained in this user's manual, then damage may occur and/or the unit may not fulfill its specifications. All non-factory repairs and/or modifications will void the warranty.
The warranty is limited to the costs of repair and/or replacement of the product. Costs for installation, labor or shipping of the defective parts are not covered by this warranty.

3 OPERATION

3.1 FEATURES

The Mastervolt PowerCharger is a fully automatic battery charger. This means that under normal circumstances it may stay switched on with the AC power and batteries connected. The PowerCharger is suitable for charging the following battery types: Lead Acid/ Wet , AGM, Spiral, Gel or Deep cycle batteries. It has an auto-ranging input facility which makes it feasible to operate on almost any AC power source in the world. It operates smoothly on both 120V/240V as well as on 230V without any compromises for the output current. The three-step Plus charging method guarantees that the batteries are always charged to 100%.
With an external AC source connected, the PowerCharger charger also serves the function of an AC to DC converter to supply DC loads which are connected to the batteries.

3.2 SWITCHING ON / STAND-BY

The PowerCharger is activated by holding the POWER switch pressed for approx. 3 seconds. The POWER switch will illuminate green. The state of charge (which is stored in the memory of the PowerCharger) will be displayed. If necessary and if AC power is available, the PowerCharger will start to charge the batteries.

The PowerCharger automatically resumes operation when reconnected to an AC-source.

By pressing the POWER button again for approx. 3 seconds, the PowerCharger will switch back to Stand-by and the POWER switch illuminates red.

WARNING
Switching the PowerCharger to “standby” does not cut off the connection to the batteries or the AC source. This means that voltages are still available inside the PowerCharger.

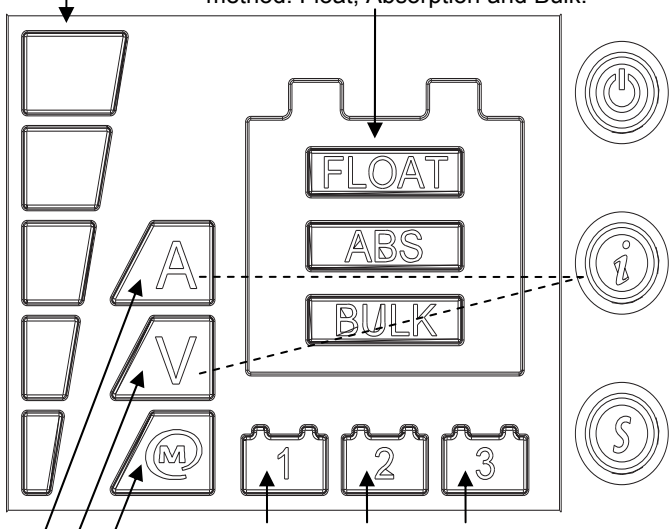
If the PowerCharger was switched to standby or AC power was lost, the POWER switch starts to blink red. After approximately 2 minutes it stops blinking and the display switches off, thus preventing battery load by the indication light.

3.3 DISPLAY

The PowerCharger is equipped with a multicolor LED display. Different LED colors and combinations have different meanings. Combination of the current display (A) with the load bar shows the percentage of maximum current, of the three battery banks together.
Combination of (V) with the load bar shows the actual charging voltage.

Meaning Load bar display

Yellow		Red
+ A	+ V	
Current 100%	>14V	Wrong AC voltage**
Current 80%	13-14V	Charger failure**
Current 60%	12-13V	Battery voltage too high**
Current 40%	11-12V	Internal temperature too high**
Current 20%	10-11V, Blinking: 10-10,5V	Battery low*, short circuit. Blinking: AC not available



POWER
Hold POWER pressed for 3 seconds to switch the charger on/ Standby. Illuminating green = on, red = standby.

INFO
Press INFO shortly to switch display: Current (A), Voltage (V).

SOURCE
Press SOURCE to select the battery bank (1, 2 or 3) you want to monitor.

Battery bank 1, 2, 3, selectable with Source button.

Current display together with load bar

Voltage display together with load bar

Illuminates: MasterBus connected

*At Battery low, the bank number concerned is blinking. Selecting another bank is still possible then, the display will return after 5 seconds.

**The Power button blinks.

Figure 2: Display operation of the PowerCharger

3.4 THREE STEP CHARGE ALGORITHM

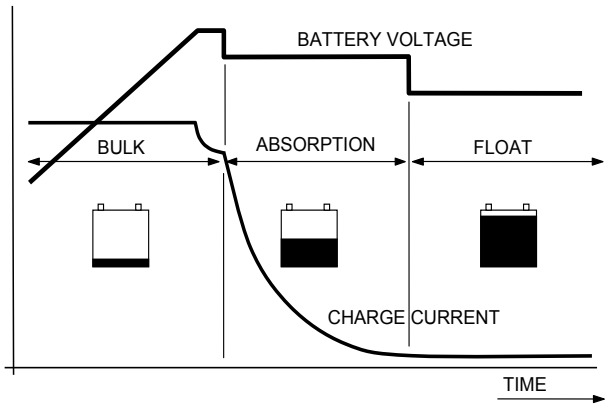


Figure 3: Three step Plus charge system

See figure 3. Battery charging is accomplished in three automatic stages: BULK, ABSORPTION and FLOAT.
The first step of the three step Plus charge system is the BULK phase, in which the output current of the charger is 100%, and the greater part of the capacity of the battery is rapidly charged. The current charges the batteries and gradually the voltage rises to the absorption voltage of 14.4V at 25°C / 77°F. The duration of this phase depends on the ratio of battery to charger capacity, and also on the degree to which the batteries were discharged to begin with.

The bulk phase is followed by the ABSORPTION phase. Absorption charging starts when the voltage on the batteries has reached 14.4V at 25°C / 77°F, and ends when the battery is completely full. Battery voltage remains constant at 14.25V at 25°C / 77°F throughout this stage, and the charge current depends on the degree to which the battery was initially discharged, the battery type, the ambient temperature, and so on. With a Lead Acid/ Wet battery this stage can last four hours, with gel and AGM approx. three hours. Once the battery is 100% full, the PowerCharger automatically switches over to the float phase.
During the FLOAT phase the PowerCharger switches to 13.25V at 25°C / 77°F and stabilizes this voltage to maintain the batteries in an optimum condition. Connected DC-loads are powered directly by the charger. If the load is higher than charger capacity, the required additional power comes from the battery, which will be progressively discharged until the charger automatically switches back to the bulk phase. Once consumption decreases, the charger goes back to normal operation of the three-step charge system.
As the PowerCharger is equipped with a three-step Plus charge system, the batteries can also remain connected to the PowerCharger during periods of unuse. One hour every 12 days the charger automatically switches to absorption to keep the battery running properly and prolong its life span. The three-step Plus charge system is also safe for all the connected equipment.

See also section 7.3 for detailed characteristics of the three step Plus charge system.

3.4.1 Temperature compensated charging

By installing the optional battery temperature sensor the charge voltage is automatically adjusted to the battery temperature.

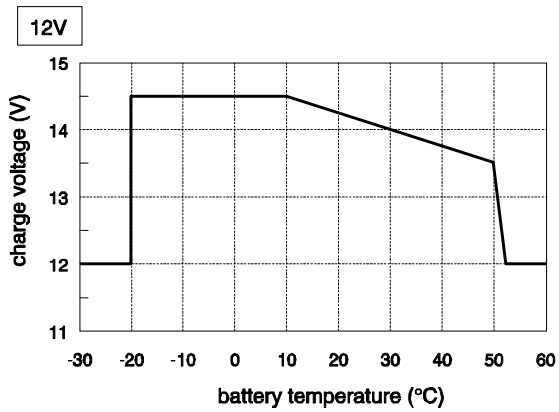


Figure 4: Temperature compensated charging

See figure 4. When the battery temperature is low, the charge voltage increases. On the other hand, when the battery temperature is high, the charge voltage is decreased. Overcharging and gassing are prevented this way. This will extend the life of your batteries.

3.4.2 Connection of a second and third battery

The PowerCharger is equipped with three equal outputs. The total output current is divided over these three outputs. See section 4.6 for connection.

3.5 MAINTENANCE

No specific maintenance to the PowerCharger is required. Examine your electrical installation on a regular basis, at least once a year. Defects such as loose connections, burnt wiring etc. must be corrected immediately.
If necessary, use a soft clean cloth to clean cabinet of the PowerCharger. Never use any liquids, acids and/or scourers.

3.6 FAILURES

The PowerCharger is protected against overload, short circuit, over heating and under and over voltage. If a fault condition occurs, a load bar segment on the display illuminates red. The LED position indicates the failure cause. See section 3.3 for explanation.

CAUTION!
The PowerCharger is not protected against:

- reversing polarity of the DC-output.
- three phase AC on the AC-input.

4 INSTALLATION

During installation and commissioning of the PowerCharger, the Important Safety Instructions in Chapter 2 are applicable at all times.

4.1 UNPACKING

In addition to the PowerCharger the delivery includes:

- a mounting bracket to mount the PowerCharger onto a wall;
- a terminating device (see chapter 6);
- the installation manual

After unpacking, check the contents for possible damage. If in doubt, contact your supplier.
Verify from the identification label (see section 1.4) whether the battery voltage is the same as the nominal output voltage of the PowerCharger (e.g. 24V battery set for a 24V battery charger).

4.2 ENVIRONMENT

The following elements must be considered during installation:

- The PowerCharger is designed for indoor use only.
- Ambient temperature: 0 ... 60°C / 32°F … 140°F; (power derating above 40°C / 104 °F to decrease the internal heat sink temperature).
- Humidity: 0-95% (non- condensing).
- Mount the PowerCharger vertically, with the connecting cables downwards.
- Make sure that the hot air that is developed during operation can be discharged. The PowerCharger must be mounted in such a way that obstruction of the airflow through the ventilation openings will be prevented.
- No objects must be located within a distance of 10 cm / 4 inches around the PowerCharger.
- Do not locate the PowerCharger in the same cabinet as the batteries.
- Do not install the PowerCharger directly above the batteries because of possible corrosive fumes.

4.3 WIRING AND BATTERIES

WARNING!
The wire and fuse sizes stated in this manual are given as a minimum. Prescribed wire and fuse sizes may be different due to local applicable regulations and standards.

4.3.1 AC wiring

For a safe installation the correct wire cross section/size must be applied. Don't use a cross section/size that is smaller than indicated. See table below to select the appropriate cross section/size for the AC wiring (up to 2m / 6ft length):

AC Current	Minimum cross section:	
	in mm²	AWG
4-6 A	1.0 mm²	17
6-12 A	1.5 mm²	15

Connection of AC-wiring and recommended wire colors

- 230V/50Hz (or 240V/60Hz) installations:

Wire color	Meaning	Connect to:
Brown or black	Phase 1	L1
Blue or red	Neutral/ Phase 2	N/ L2
Green/Yellow	Ground	PE / GND

- 120V/60Hz installations (single phase):

Wire color	Meaning	Connect to:
Black	Hot or Line	L1
White	Neutral	N
Green	Ground	PE / GND

4.3.2 DC wiring

Keep in mind that high current will pass through the DC wiring. Keep the cable length as short as possible, this will keep the system efficiency as high as possible. The recommended minimum cross section of maximum 3m/ 9ft length battery cables for output 1, 2 and 3 is:

Model PowerCharger	DC Cable cross section:
12/40-3	16.0 mm² / 4 AWG

Use ring terminals on the ends of the wires. The terminals must be crimped with a proper crimping tool. Use the following wire colors for DC wiring color to make a clear distinction between the positive and negative wire from the battery:

Wire color	Meaning	Connect to:
Red	Positive	+ (POS)
Black or Yellow	Negative	– (NEG)

Lay the positive and negative cables next to each other to limit the electro magnetic field around the cables. The negative cable should be connected directly to the negative post of the battery bank or the ground side of a current shunt. Do not use the chassis frame as the negative conductor. Tighten securely. The positive battery cable must be fused and connected to the positive post of the battery bank.

The recommended cable fuse for outputs 1, 2 & 3 is 50 A.

The fuse with the fuse-holder is available from your local Mastervolt distributor or Customer Service Representative, see chapter 9 Ordering Information.

4.3.3 Battery capacity

The minimum required battery capacity is 80 Ah.

4.3.4 AC safety grounding

WARNING!
The ground wire offers protection only if the cabinet of the PowerCharger is connected to the safety ground. Connect the ground terminal (PE / GND) to the boats grounding system or the vehicles chassis.

4.4 OVERVIEW CONNECTION COMPARTMENT

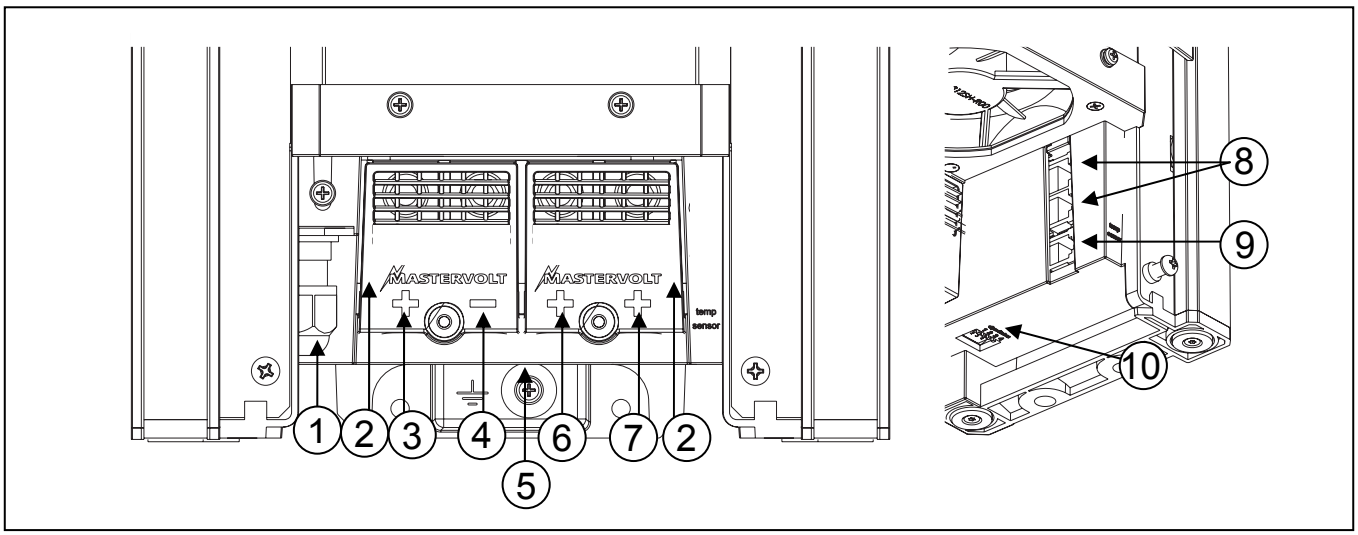


Figure 5: Overview connection compartment

1. Cable gland for AC-wiring
2. Isolation caps for DC connections
3. Positive terminal charge output 1
4. Common negative output terminal
5. Ground stud
6. Positive terminal charge output 2
7. Positive terminal charge output 3
8. Communication connectors
9. Temperature sensor jack
10. DIP switches

4.5 THINGS YOU NEED

Make sure you have all the parts necessary to install the PowerCharger:

	Quantity
PowerCharger (included)	1
Battery temperature sensor with cable and plug (optional).	1
DC cable to connect the positive DC connection (+) of the PowerCharger to the positive pole of the DC-distribution; for specifications see section 4.3.2.	1
DC-cable to connect the negative DC connection (–) of the PowerCharger to the negative pole of the DC-distribution; for specifications see section 4.3.2.	1
DC-fuse holder with a DC-fuse, to be integrated in the positive DC-cable. For specifications see section 4.3.2.	1
Screws / bolts (Ø 6mm) (with plugs) to mount the enclosure to a surface. Use mounting materials which are suitable to carry the weight of the PowerCharger.	4
AC cable* to connect the AC input to an external power source (e.g. a shore connection or a generator set);	1
Batteries. See section 4.3.3 for recommended capacity	X
Appropriate and reliable cable terminals, cable lugs, battery terminals and cord end terminals	X

* Insulated three wire cable with wire colors according to the local applicable regulations. The applicable length and wire diameter depend on the electrical installation (see section 4.3.1).
See chapter 7 Ordering Information.

4.6 CONNECTIONS

WARNING
Installation must be performed by qualified (marine) electricians. Before beginning with the connection of the wiring, ensure there is no power on the AC or DC systems. Verify there is no voltage present using a volt meter or test apparatus.

CAUTION!
Short circuiting or reversing polarity may lead to serious damage to the batteries, the PowerCharger, the cabling and/or the terminal connections. Fuses between the batteries and the PowerCharger can not prevent damage caused by reversed polarity. The damage as a result of reverse polarity is detectable by the service department and is not covered by the warranty.

CAUTION!
Undersized cables and/ or loose connections can cause dangerous overheating of the cables and/or terminals. Therefore
use cables of the correct size and tighten cables properly and securely.

NOTE:
If the battery temperature remains within 15-25°C (59-77°F), connection of the battery temperature sensor is optional.

4.6.1 Connection example

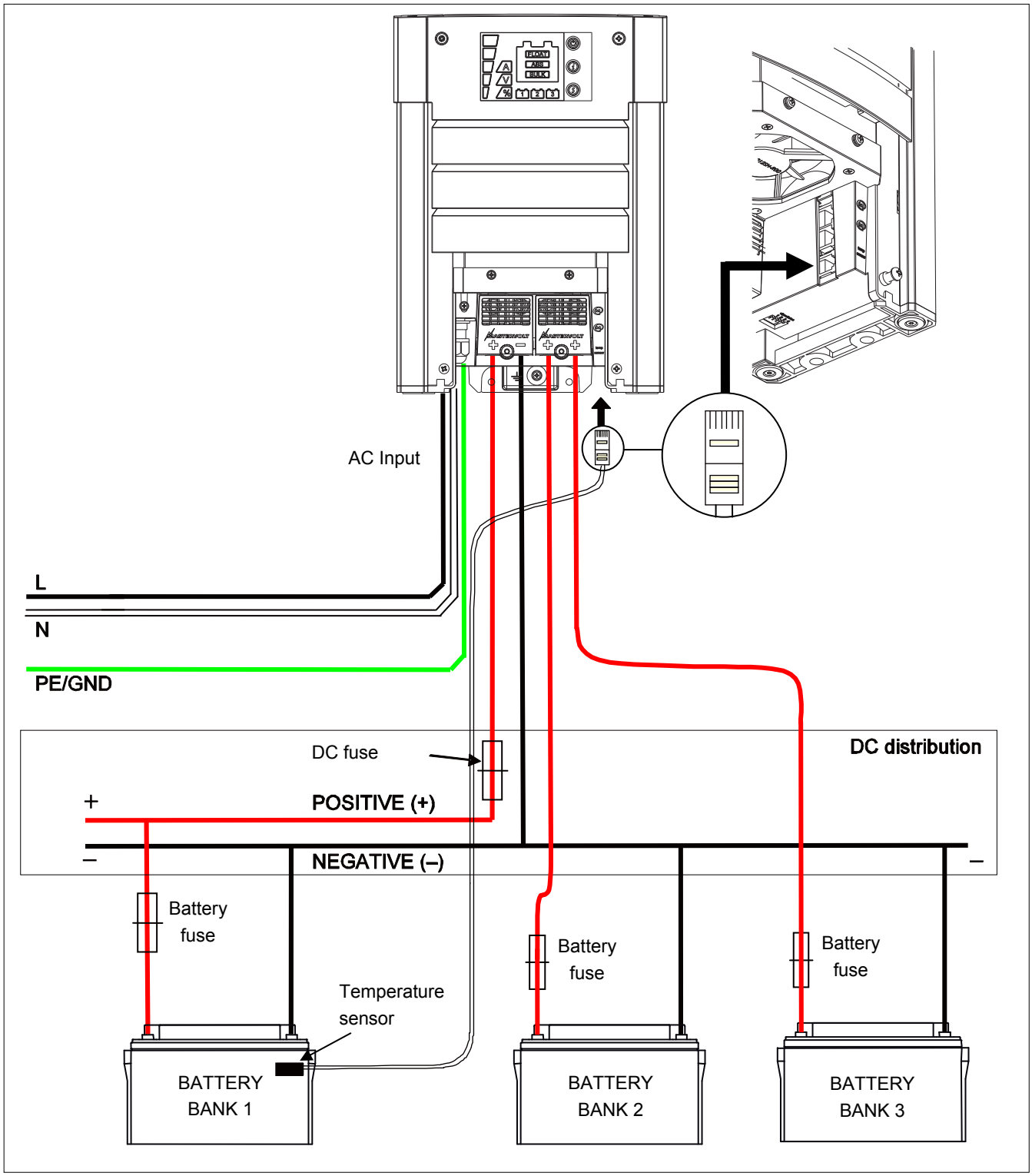


Figure 6: installation example of the PowerCharger

This schematic is to illustrate the general placement of the PowerCharger in a circuit. It is not meant to provide detailed wiring instructions for any particular electrical installation.

4.7 COMMISSIONING AFTER INSTALLATION

If your PowerCharger is not new, you have to take into account that former users may have changed the settings. Refer to chapter 5 for settings.

4.7.1 General

The factory settings of the PowerCharger are optimal for most installations. With some applications however, it is desirable to change these settings. Therefore several adjustments can be made. See chapter 5.

NOTE:
The DIP-switches must be adjusted *prior* to commissioning; all other settings can only be made *after* commissioning.

CAUTION!
Check the polarity of all wiring before commissioning: positive connected to positive (red cables), negative connected to negative (black cables).

If all wiring is OK, place the DC-fuse(s) of the DC-distribution to connect the batteries to the PowerCharger.

WARNING
When placing this fuse, a spark can occur, caused by the capacitors used in the PowerCharger. This is particularly dangerous in places with insufficient ventilation, due to the gassing of the batteries an explosion can occur. Avoid having flammable materials close by.

Now the PowerCharger is ready for operation. After switching on the AC power supply the PowerCharger will initiate the charging process.

4.7.2 Communication network

For remote display of battery charger information, the Masterview Read Out Remote Panel (optional) may be purchased. During first commissioning the PowerCharger will be recognized by the Read Out Panel automatically.
Refer to the MasterView Read Out user's manual for information on how to install the panel.

4.8 DECOMMISSIONING

If it is necessary to put the PowerCharger out of operation, follow the instructions in order of succession as described below:

1 Switch the PowerCharger to stand-by (section 3.2).
2 Remove the DC-fuse(s) of the DC-distribution and/or disconnect the batteries.
3 Remove the AC-fuse(s) of the AC-input and/or disconnect the AC-mains or dedicated breaker.
4 Open the connection compartment of the PowerCharger.
5 Check with a suitable voltage meter whether the inputs and the outputs of the PowerCharger are voltage free.
6 Disconnect all the wiring.

Now the PowerCharger can be demounted in a safe way.

4.9 STORAGE AND TRANSPORTATION

When not installed, store the PowerCharger in the original packing, in a dry and dust free environment.
Always use the original packing for transportation. Contact your local Mastervolt Service Center for further details if you want to return the apparatus for repair.

4.10 RE-INSTALLATION

To reinstall the PowerCharger, follow the instructions in this chapter (4).

5 DIP SWITCH SETTINGS

Adjustments to the PowerCharger can be made by means of DIP-switches:

DIP-switch	Description	Default
1. Battery type	AGM/Gel batteries. Off = Lead Acid/ Wet , On = AGM/Gel	Off
2. Charge algorithm	12V power supply. Off=disabled, On=enabled	Off
3. Display mode	Display power safe mode. Off=disabled, On=enabled	Off
4. Equalize mode	Equalize is suitable FOR LEAD ACID/ WET BATTERIES ONLY!. Off=disabled, On=enabled	Off

CAUTION!
Incorrect settings of the PowerCharger can cause serious damage to your batteries and/or the connected load! Adjustments of settings should be undertaken by authorized personnel only.

5.1 DIP SWITCH SETTINGS

The PowerCharger has four DIP switches at the bottom side of the housing. See figure 7. DIP switch 3 is without function.

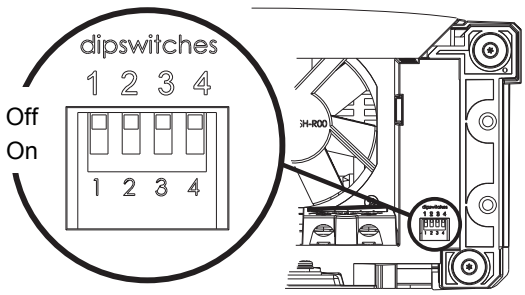


Figure 7: DIP switches

5.1.1 DIP switch 1: Battery type

The factory default for the battery type is optimal for most installations. In some applications however, it is desirable to change this setting. For special batteries, refer to the corresponding user's manual.

Lead acid (factory setting)	OFF
Gel / AGM / spiral battery (see specifications)	ON

5.1.2 DIP switch 2: Charge algorithm

IUoUo, fully automatic / 3step+ (factory setting)	OFF
Constant voltage charging (Float = 13.25/26.5V)	ON

5.1.3 DIP-switch 3: Stand-by mode for display

Display will switch off when PowerCharger is switched to stand-by mode (factory setting)	OFF
Display will stay on when PowerCharger is switched to stand-by mode. Note that the display is powered by the batteries.	ON

5.1.4 DIP switch 4: Equalize mode

WARNING
Incorrect use of the equalize mode may lead to hazardous situations.
Do not smoke; use any open flame or other sources of ignition due to risk of explosion. Always ventilate the room were the batteries are located and equalized to clear the air.
Equalization is ONLY applicable for lead acid type batteries and will damage gel and AGM type batteries.

An equalizing charge can be necessary after very deep discharges and/or inadequate charges. This has to be carried out according to the specifications of the manufacturer of the batteries.

During equalizing the batteries are brought into the gas state and voltages in excess of the permitted voltages is possible. Therefore appropriate measures must be taken, e.g. disconnect all loads from the battery and ventilate the room. For these reasons, use of the equalizing mode should only be executed by trained technicians.

The equalizing mode can only be started when the PowerCharger is in operation and in Float. This means that the connection compartment of the PowerCharger must be opened while connected to the mains and the batteries. Take adequate measures for protection against short circuits and electrical shocks.

To start the equalize mode: move DIP switch 4 to the ON position and back to the OFF position.

6 TROUBLESHOOTING

If you cannot solve a problem with the aid of this chapter, contact your local Mastervolt Service Center. See www.mastervolt.com. Make sure you have the article number and serial number available if you need to contact your Mastervolt Service Center to solve an issue. Please refer to Section 1.4 to find this information.

6.1 FAULT FINDING TABLE

Malfunction	Possible cause	What to do
No output voltage and/or current	No AC-input	Check AC wiring, check remote control panel.
	AC-input voltage too low (< 90VAC)	Check input voltage, check generator.
	AC input frequency out of range	Check input voltage, check generator.
Output voltage too low, charger supplies maximum current	Load that is connected to the batteries is larger than charger can supply.	Reduce load taken from the batteries.
	Batteries not 100% charged	Measure battery voltage. After some time this will be higher.
Charge current too low	Batteries almost fully charged	Nothing, this is normal when the battery is almost fully charged.
	High ambient temperature	Nothing; if ambient temperature is more than the setting limit the charge current is automatically reduced.
	Low AC input voltage. At lower AC input voltages the charge current is reduced. See figure 15.	Check AC input voltage.
Batteries not fully charged	Charge current too low	See “Charge current too low” in this table.
	Current to load is too high	Reduce load taken from the batteries.
	Charge time too short	Use a battery charger with higher capacity.
	Battery temperature too low	Use the battery temperature sensor (optional).
	Defective or old battery	Check battery and replace if necessary.
Batteries are discharged too fast	Battery capacity reduced due to sulphation or stagnation	Charge and recharge a few times, this might help. Check battery and replace if necessary.
Batteries are too warm, gassing	Defective battery (short circuit in cell)	Check battery and replace if necessary.
	Battery temperature too high	Use the battery temperature sensor (optional).
	Charge voltage too high	Check settings (see section 5.1).

7 TECHNICAL DATA

7.1 SPECIFICATIONS

Model	**12/40-3**
UPC code	852968002301
Article no.	44210400
Nominal input voltage:	120/240V, 230V
Nominal input frequency:	50/60Hz
Full load consumption:	700VA
Efficiency at full load (230V AC):	≥81% @ 230V input
Nominal output voltage:	12V
Total maximum charge current*:	40A at 14.4V
Number of battery outputs:	3
Charge characteristic*:	IUoUo, automatic, three step plus
Charge voltage Bulk*:	14.4 V
Charge voltage Absorption*:	14.25 V
Charge voltage Float*;	13.25 V
Max. absorption and max. bulk timer*:	8 hours (start max. bulk timer at 13.25V)
Minimum absorption time*:	15 min.
Battery type settings*:	Lead Acid/ Wet, gel, traction, AGM, spiral (adjustable by means of DIP-switch)
Dimensions in inch:	11.5x8.3x5.2
Weight:	9 Lbs
Recommended battery capacity:	80-400Ah
Power factor regulations	≤ 0.99
Temperature sense	Optional battery temperature sensor with 19 or 49 ft cable
Voltage compensation	Yes, automatic compensation with optional battery temperature sensor
DC consumption	<5mA
Temperature range	–13°F to 140°F
Cooling	Vario fan and natural cooling to ensure optimized cooling
Sound level	<52dBA / 1m
Protection degree	IP23
Approvals	Fully CE and E-marking according to automotive directive 95/54/EG Meets UL1236 including marine supplement and ABYC standards
Communication	Via MasterView Read Out Remote Panel (optional)
Communication connectivity	To MasterView Read Out Remote Panel only

* Adjustable, see chapter 5 for settings.
Specifications are subject to change without prior notice.

7.2 DIMENSIONS

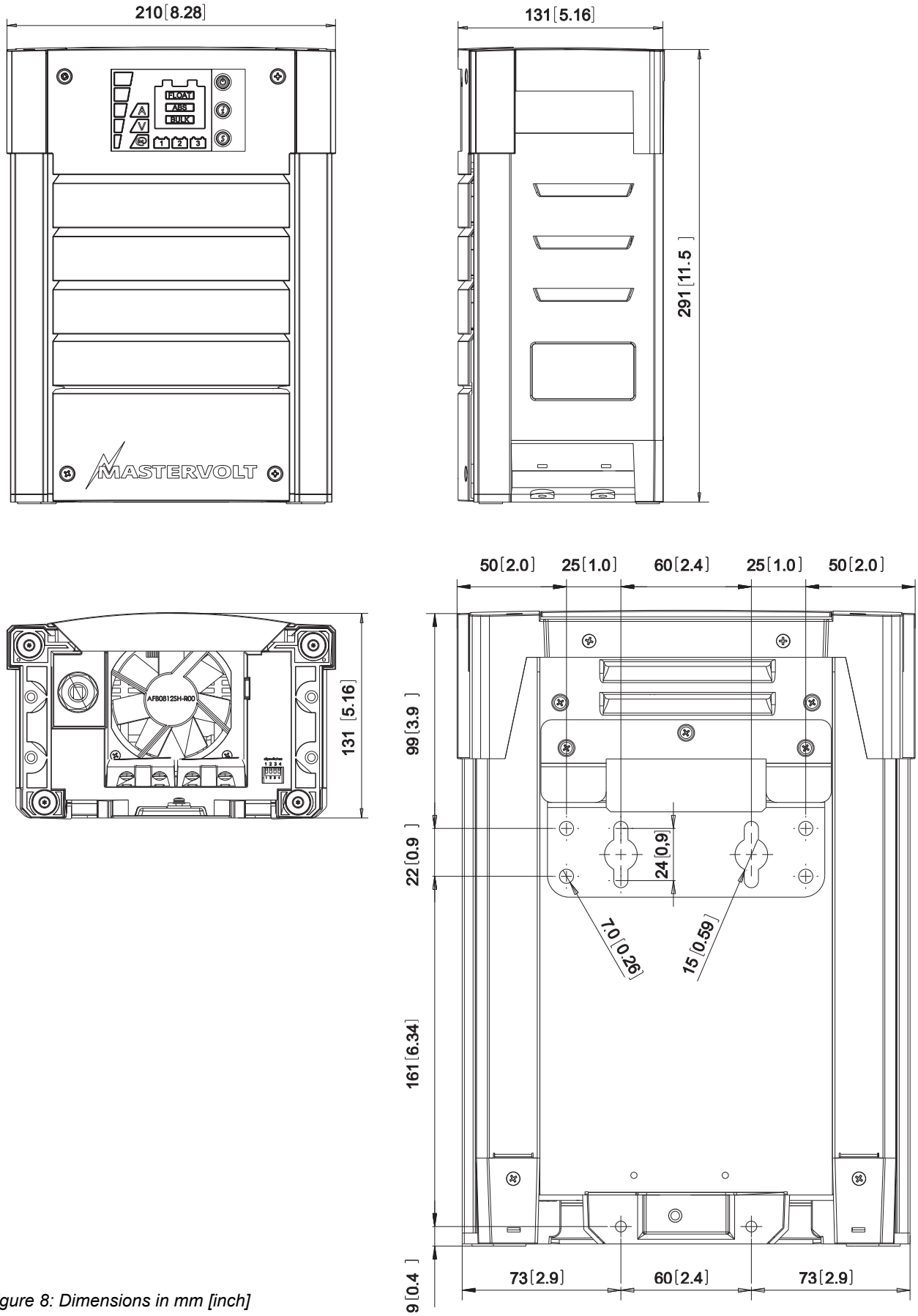


Figure 8: Dimensions in mm [inch]

7.3 CHARACTERISTICS

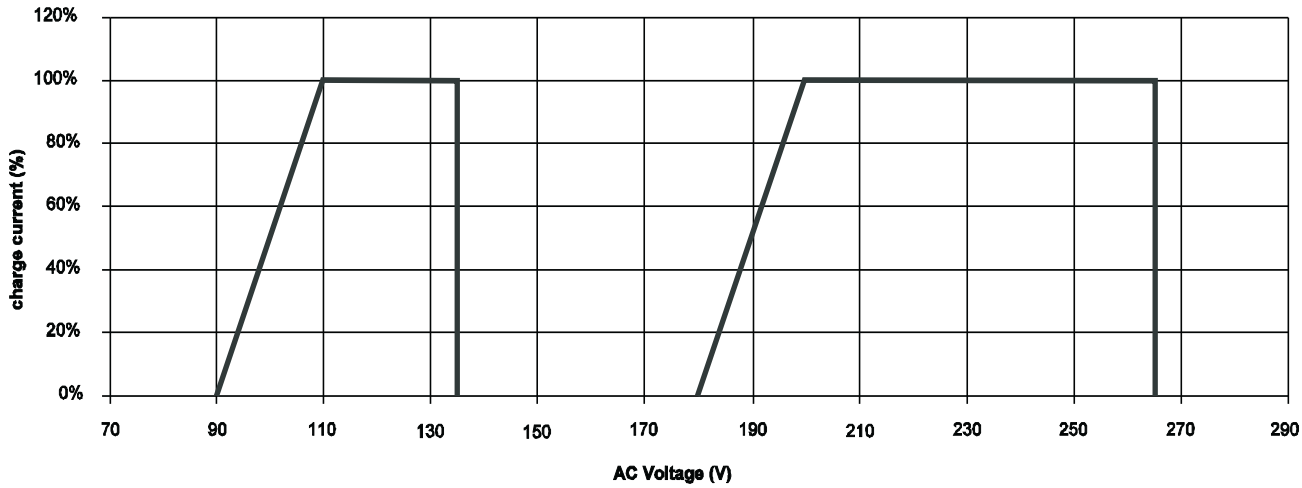


Figure 9: Charge current versus input voltage

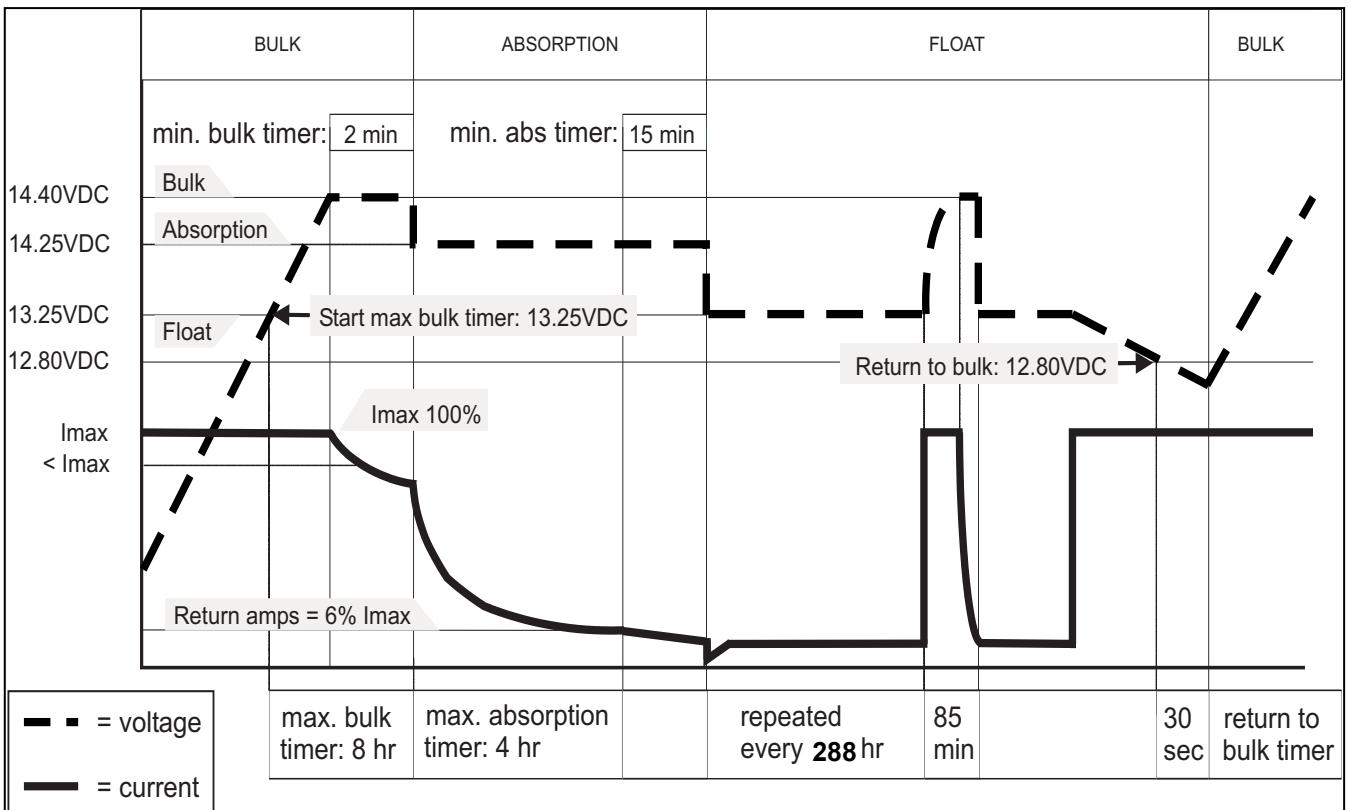


Figure 10: Charge characteristic of the three-step Plus charging method (at 25°C / 77°F)

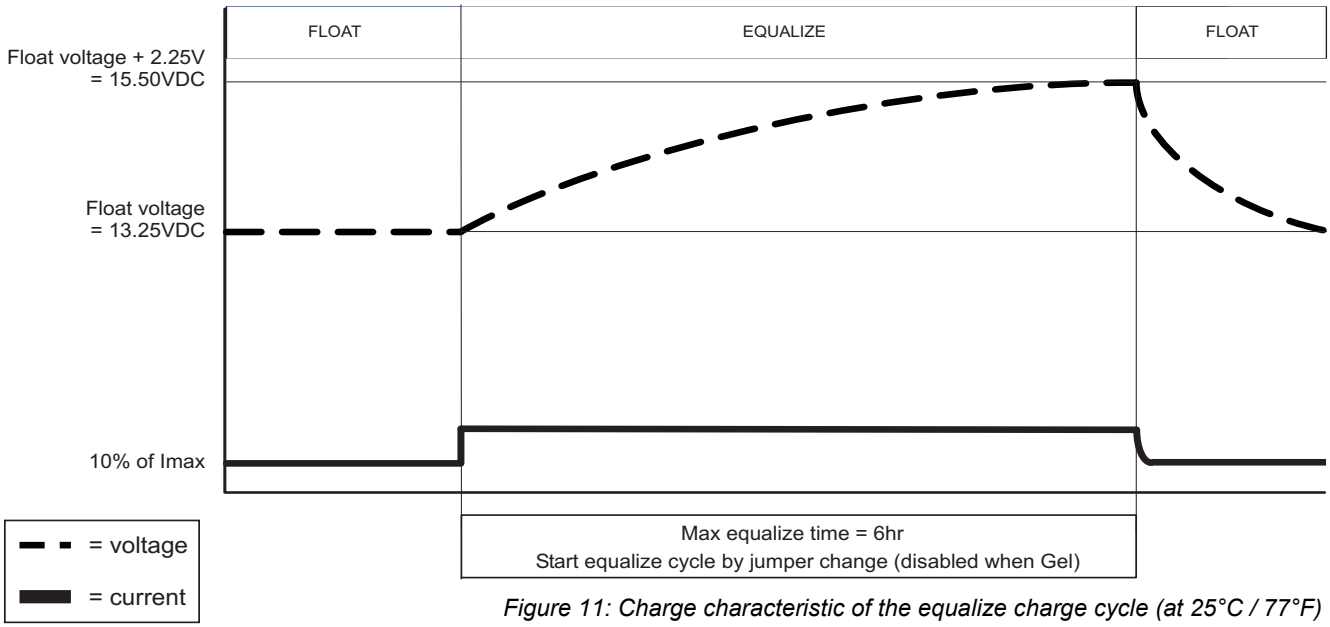


Figure 11: Charge characteristic of the equalize charge cycle (at 25°C / 77°F)

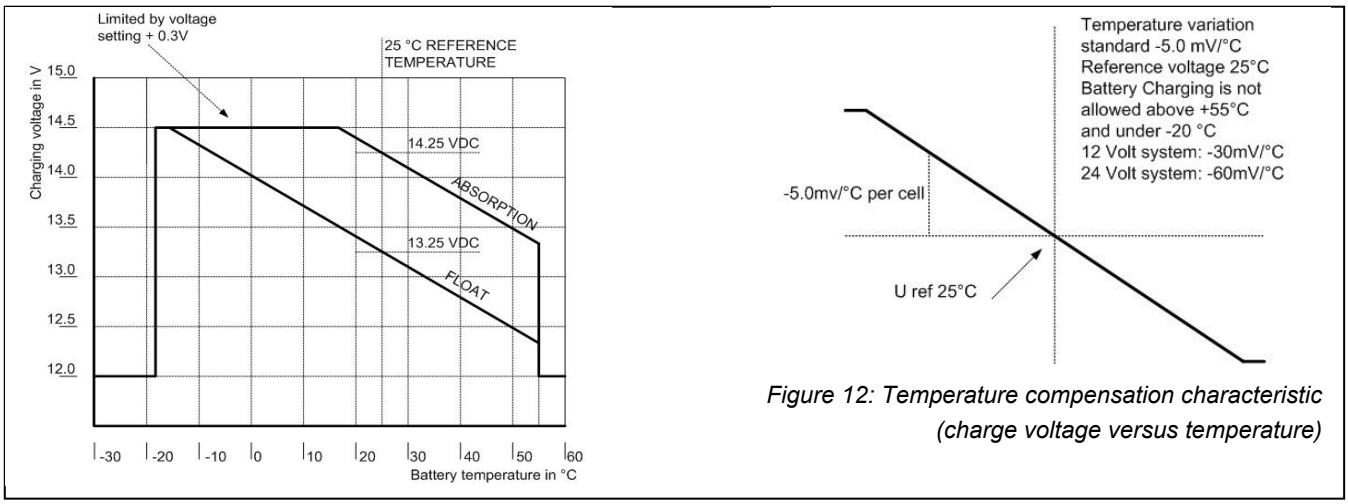


Figure 12: Temperature compensation characteristic (charge voltage versus temperature)

8 ORDERING INFORMATION

Part number	Description
79009006	Battery switch 250A with knob
79009005	Battery switch 250A with key
41500500	Battery temperature sensor, incl. 6 metres / 19 ft cable
41500800	Battery temperature sensor, incl. 15 metres / 49 ft cable
77010050	MasterView Remote Read Out panel
77010020	MasterView Remote Read Out OEM
77040000	Terminator

Mastervolt offers a wide range of products for your electrical installation, AGM, gel and Li-ion batteries, shore power connections, DC distribution kits and much more.

9 EC DECLARATION OF CONFORMITY

Manufacturer Mastervolt
Address Snijdersbergweg 93
1105 AN Amsterdam
The Netherlands

Herewith declares that:

Product:
44210400 PowerCharger 12/40-3

Is in conformity with the provision of the EC, EMC directive 2004/108/EC,
Low Voltage directive 2006/95/EC

The following harmonized standards have been applied:

EN 55022:2006, EN 55024:1998+A1:2001+A2:2003
EN 61000-3-2:2006, EN61000-3-3:1995+A1:2001+A2:2005
IEC 61000-4-2:1995+A1:1998+A2:2000, IEC 61000-4-3:2006, IEC 61000-4-4:2004, IEC 61000-4-5:2005,
IEC 61000-4-6:2007, IEC 61000-4-8:1993+A1:2000, IEC 61000-4-11:2004

EN 60950-1, EN 55022: 1998+A1: 2000+A2: 2003 Class A
EN 61000-4-2: 2001, EN 61000-4-4: 2004, EN 61000-4-5: 2001,
EN 61000-4-3: 2002+A1: 2002, EN 61000-4-6: 1996+A1: 2000,
EN 61000-3-2: 2000, EN 61000-6-4: 1997,IEC 61000-4-2 :1995+A1 :1998+A2 :2000

Amsterdam,

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